Hybrid State Space Models for Long Sequences

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ABSTRACT

An empirical study evaluating selective state space models against quadratic attention mechanisms.

1. Introduction & Research Objectives

This research investigates Hybrid State Space Models for Long Sequences. The primary research objective is to advance state-of-the-art methodology in machine learning, offering theoretical justifications and empirical validation across standard benchmarks.

2. Methodology & Proposed Architecture

Hardware-aware selective scan kernels with SRAM cache tiling.

3. Empirical Results & Comparative Evaluation

Achieves linear compute complexity O(N) with 4.5x faster generation throughput.

4. Disclosed Limitations & Boundary Conditions

Lack of associative memory on multi-hop associative recall tasks.

5. Future Research Directions & Open Inquiries

Future trajectories focus on sub-quadratic attention scaling, automated hyperparameter optimization, and extending the methodology to multi-modal visual and auditory reasoning.